Status	Finished
Started	Saturday, 24 October 2026, 10:57 PM
Completed	Saturday, 24 October 2026, 11:07 PM
Duration	9 mins 35 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search can find multiple instances of the target element
d. Binary search works on unsorted lists

Question **2**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list contains duplicate elements
b. When the element is found in the list ✓
c. When the list is sorted
d. When the search algorithm finishes

Question **3**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94
b. 90 and 100
c. 94 and 99 ✗
d. 90 and 99

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. The first element is returned
c. An error is raised
d. The search is considered unsuccessful ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search stops immediately ✓
b. The search continues until the end of the list
c. The search backtracks to find duplicate elements
d. The search starts over from the beginning

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search requires the list to be sorted.

Question **7**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Insertion

c. Bubble ✓

d. Complexity

Question **8**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Determining whether an element is present in a list ✓

b. Deleting elements from a list

c. Sorting elements in a list

d. Inserting elements into a list

Question **9**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Binary search

b. Jump search

c. Linear search ✓

d. Hashing

Question **10**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search ends successfully ✓
c. The search continues in the right sublist
d. The list is sorted

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

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b. The list must be sorted
c. There should be direct access to the middle element in any sublist
d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **12**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000
b. 100
c. 500
d. 10 ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓
b. It is slower than linear search for small lists
c. It does not work with negative numbers
d. It can only be applied to large lists

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 90 and 94
c. 89 and 94
d. 90 and 99 ✓

Question **15**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element
b. The middle element is chosen randomly
c. The search stops
d. The lower middle element is chosen as the middle element ✗